

TabM-UQ: Uncertainty-Aware Tabular Deep Learning via Parameter-Efficient Ensembling

- Single model: default TabM, 32 sub-models, no embeddings (paper default)
- First uncertainty + within-dataset OOD evaluation of TabM (ICLR 2025 future work)
- 11 benchmarks x 3 seeds = 33 runs, CPU-only, about 30 hours total
- Team: [Your Name], [Roll No] | Guide: [Guide Name] | Batch: [9A]
- Covers Review 1 (Acceptance) + Review 2 (Design / Phase-1 gate)

Cover - replace bracketed placeholders before submission

R1. Problem Statement (Thesis 1.2)

- Tabular models lack reliable uncertainty estimates for high-stakes use
- TabM gives 32 predictions per object, but their meaning was never evaluated
- Silent failures possible in medical, credit and fault-detection domains
- Paper leaves uncertainty/OOD as explicit future work; still unaddressed

Thesis Sec 1.2 - text-only slide (no picture needed)

R1. Motivation and Need (Thesis 1.3)

- TabM (ICLR 2025): state-of-the-art tabular deep learning, used in Kaggle wins
- Gradient-boosted trees are accurate but provide no uncertainty
- Gap is explicit and open; a CPU-only study is feasible
- 'I don't know' is more valuable than a wrong confident prediction

R1. Project Objectives (Thesis 1.4)

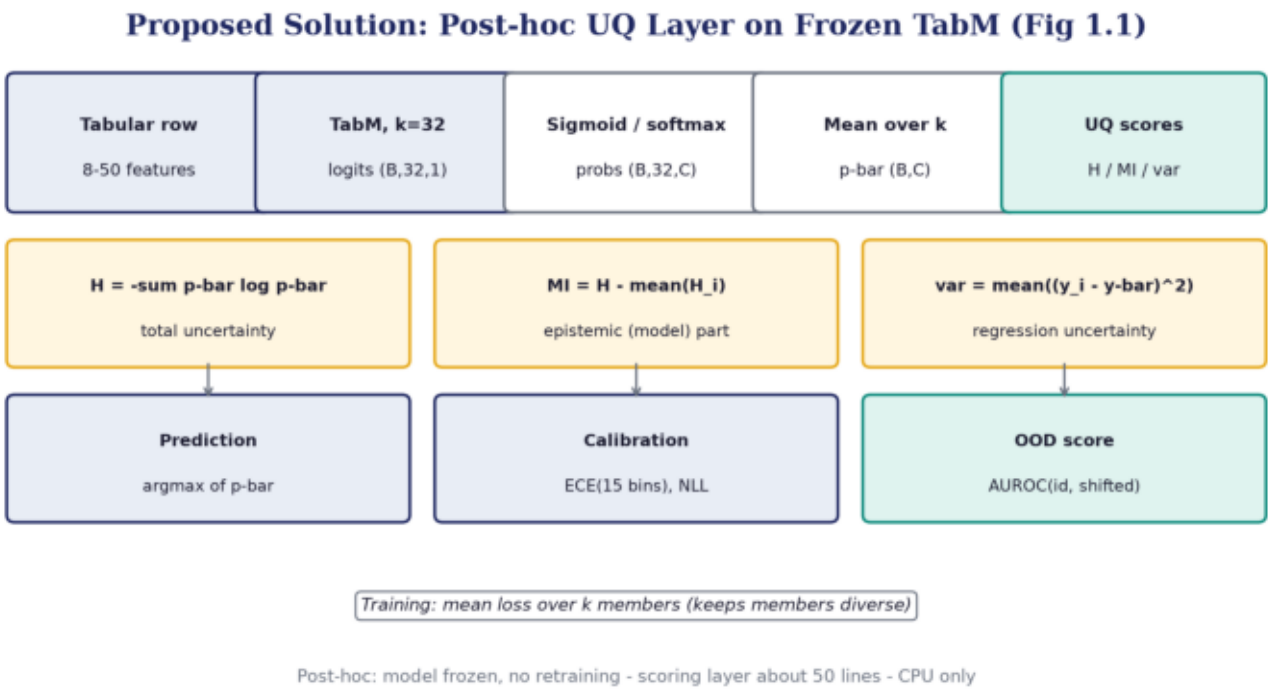
- O1: reproduce default TabM on 11 benchmarks (33 runs)
- O2: quantify uncertainty - entropy, MI (classification) + variance (regression)
- O3: detect OOD within-dataset - noise / mask / shuffle / scale (7 shifts)
- O4: calibrate - ECE (15 bins), NLL + reliability diagrams

Thesis Sec 1.4, Table 1.4 - type as Word table in thesis

R1. Proposed Solution Overview (Thesis 1.5, Fig 1.1)

- Frozen TabM (B,32,1) + post-hoc scoring layer (about 50 lines)
- Training uses mean loss over members, not loss of the mean (keeps diversity)
- Outputs: prediction + uncertainty + OOD score; no retraining, CPU only

Fig 1.1: detailed UQ-layer pipeline with shapes and formulas



R1. Scope and Boundaries (Thesis 1.6, Table 1.1)

- IN: vanilla 32-sub-model TabM, 11 datasets, entropy/MI/variance, ECE/NLL, 7 shifts
- OUT: embedding variant, tuning, external baselines, cross-dataset shifts
- Baselines deferred - workshop target needs uncertainty + OOD only

Thesis Sec 1.6, Table 1.1 - type as Word table in thesis

R1. Technology Stack (Thesis 3.8)

- Python 3.11, PyTorch 2.12 (CPU), official TabM package 0.0.3
- scikit-learn 1.9 (encoding + metrics), matplotlib 3.10 (plots)
- Data: official 50-set TabM suite, about 1 GB, 11 sets used
- Compute: 3 parallel workers x 5 threads, no GPU required

Thesis Sec 3.8, Table 3.8 - type as Word table in thesis

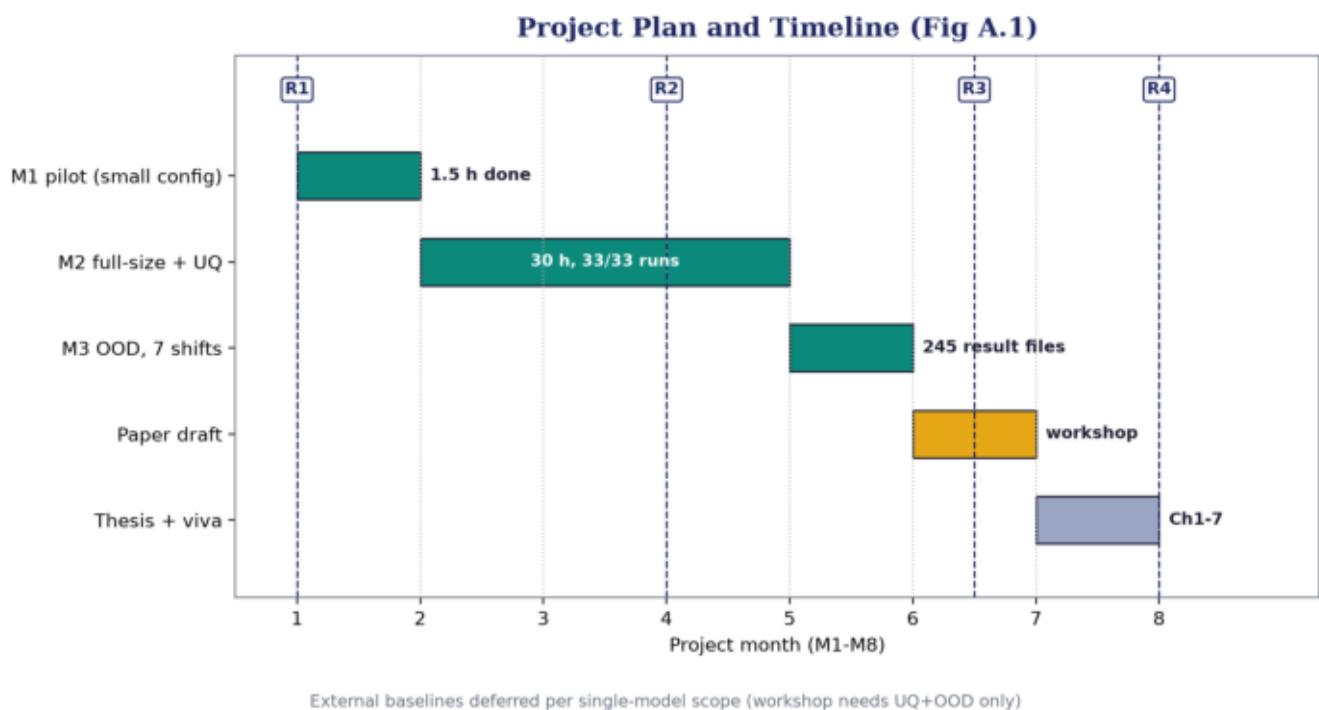
R1. Literature Survey (Thesis Ch 2, Table 2.1)

- [1] TabM: best tabular accuracy, no uncertainty/OOD - our starting point
- [2] Deep ensembles: uncertainty gold standard, needs 32 training runs
- [3] BatchEnsemble: efficient ensembling idea behind TabM
- [4] MC Dropout: cheap single-model uncertainty, weaker calibration
- [5] Softmax OOD baseline: standard AUROC protocol
- [6] Calibration (ECE/NLL): our calibration metrics
- Gap: no published work evaluates TabM outputs as uncertainty/OOD scores

R1. Project Plan and Timeline (Appendix A, Fig A.1)

- M1 pilot (small config, 1.5 h) done | M2 full-size + UQ (30 h, 33/33) done
- M3 OOD over 7 shifts (245 files) done | Paper: workshop target
- Reviews: R1 acceptance - R2 gate - R3 paper - R4 final

Fig A.1: Gantt chart with measured runtimes and review gates



R1. Team Roles and Report Organisation (Thesis 1.8)

- Solo project: data loading, model wrapper, metrics, OOD, results by one owner
- Thesis Chapters 1-7 plus Appendices A-F; fill name/roll/guide/batch
- Fill placeholders on the PPT title slide before the review

Thesis Sec 1.8, Table A.4 - type as Word table in thesis

R2. R1 Feedback - Actions Taken (Table A.2)

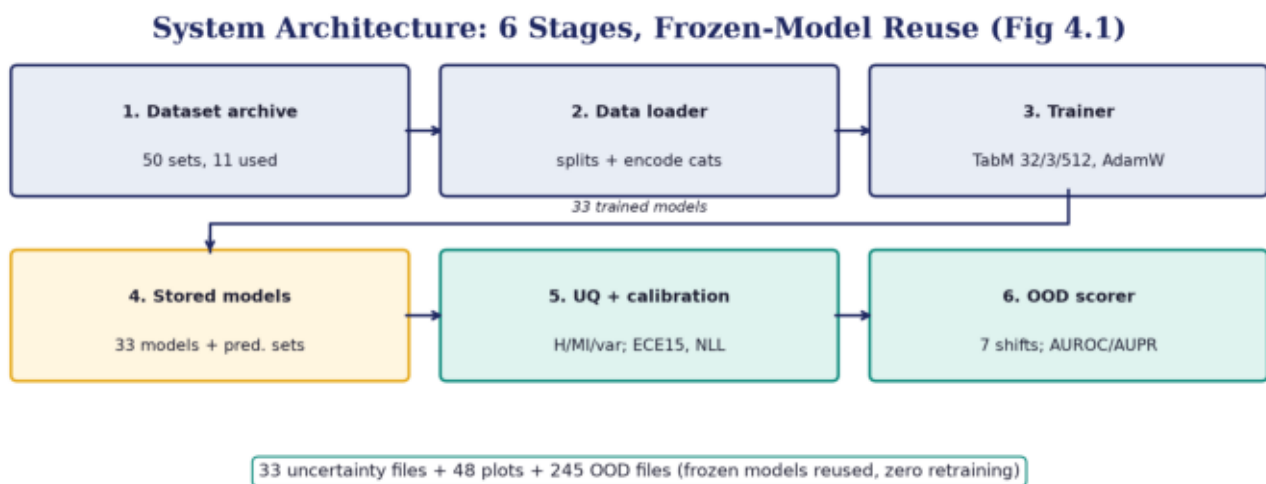
- Scope fixed to single vanilla TabM (32 sub-models, no embeddings)
- Configuration corrected to paper defaults; runtime corrected to 30 h measured
- All documents reconciled to one configuration; baselines formally deferred

Thesis Table A.2 - evidence points to thesis sections and logs

R2. System Architecture - Final (Fig 4.1)

- 6 stages: archive - loader - trainer - stored models - UQ/calibration - OOD
- 33 trained models reused frozen for all analysis (zero retraining)
- Outputs: 33 uncertainty files, 48 plots, 245 OOD files

Fig 4.1: six stages with inputs, outputs and artifact counts



Early stopping (patience 20) - batch 256-2048 - seeds 0,1,2 - CPU only

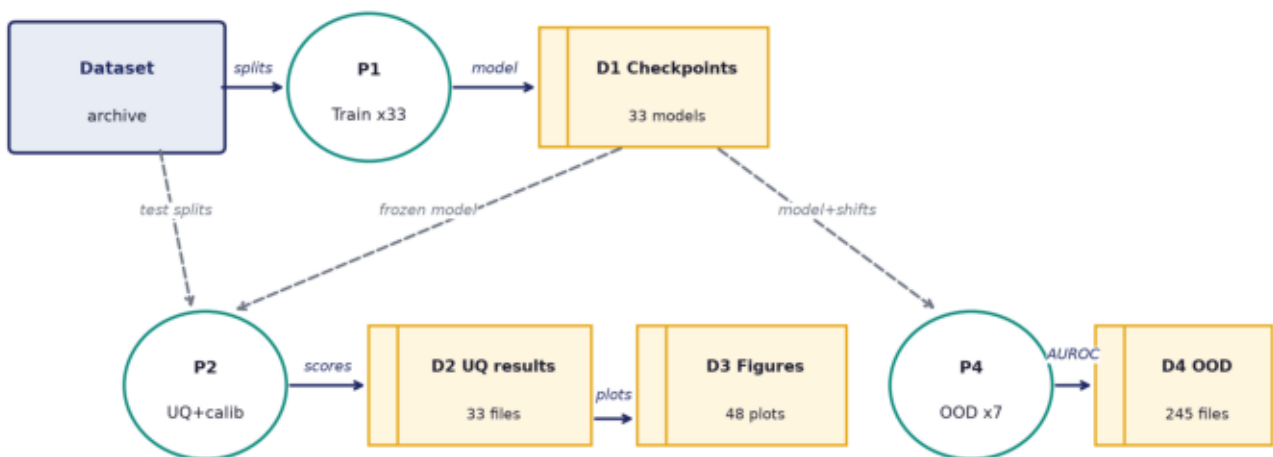
Binary indicator features of 2 sets excluded (identical features for ID and OOD)

R2. Data Flow and Interface (DFD Levels 0-1)

- P1 trains 33 models from dataset splits into the checkpoint store
- P2 scores uncertainty + calibration (test splits + frozen model)
- P4 scores OOD over 7 shifts; results to dedicated stores
- No database: versioned file archive; 3 command-line operations

DFD: external entity, processes P1/P2/P4, stores D1-D4

Data-Flow Diagram, Levels 0-1 (Processes, Stores, Flows)



No database: versioned file archive. Interface: train / evaluate-UQ / evaluate-OOD commands.

R2. Module Implementation (Thesis 5.3)

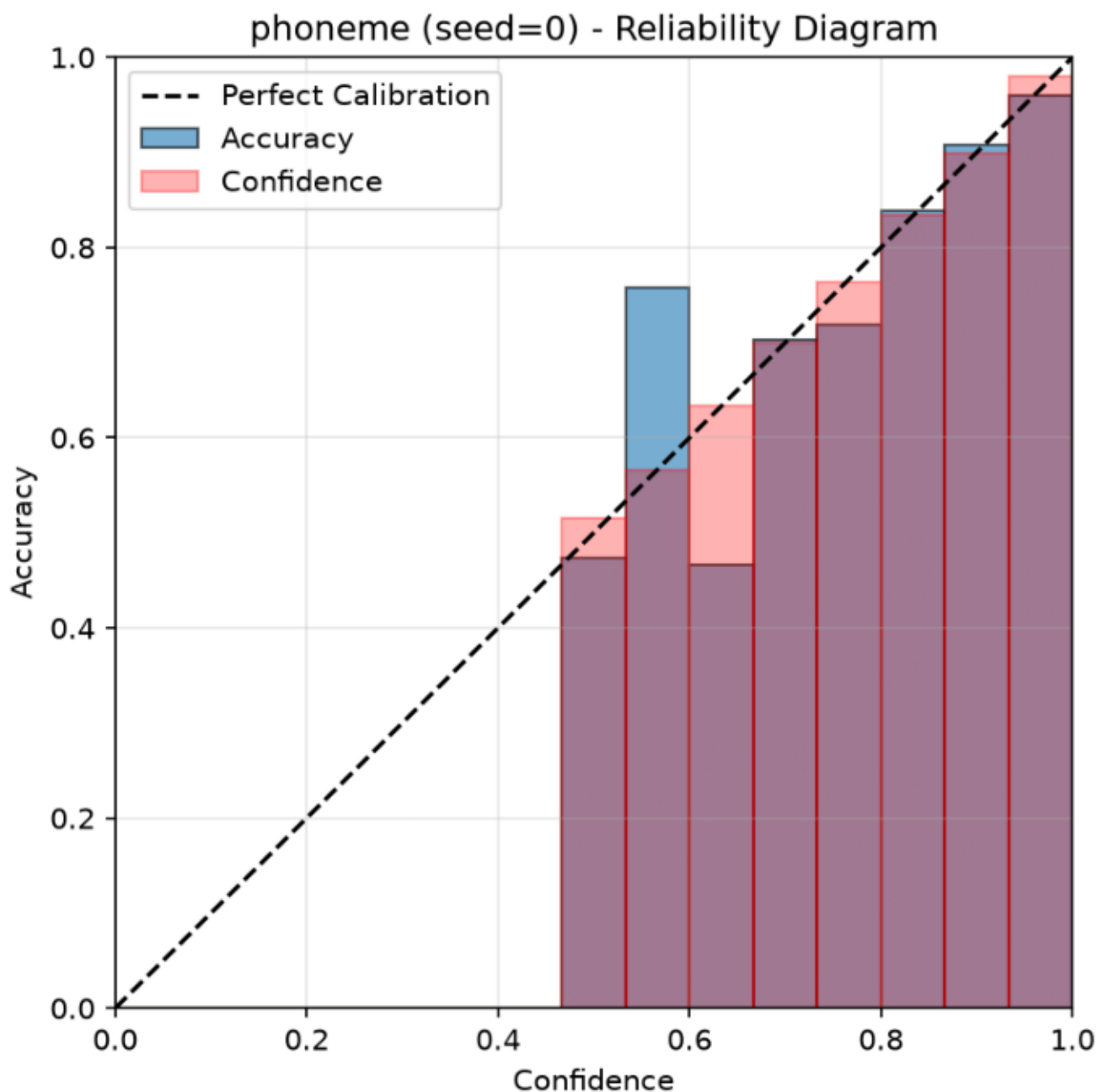
- Data loader: reads splits, encodes string categories, reserves unknown code
- Model wrapper: TabM with 32 sub-models, AdamW, auto batch, early stopping
- Scorers: sigmoid to probabilities, entropy/MI/variance, ECE/NLL, AUROC
- OOD: seeded corruptions (noise/mask/shuffle/scale); per-seed result files

Thesis Sec 5.3 - detailed bullets, key listings in Appendix E

R2. Live Demo - Full System (Fig 5.2)

- Phoneme dataset: load - predict - entropy 0.324, ECE 0.024
- Reliability diagram and entropy histogram shown below
- OOD on phoneme: mask-0.5 AUROC 0.564 (entropy histogram in folder)

Fig 5.2: phoneme reliability diagram (live demo backup)



R2. Testing Results (Tables 6.3-6.5)

- Unit: wrapper/metrics/calibration imports - pass
- Integration: uncertainty recompute 33/33 with matching summaries - pass
- System: OOD 7 shifts x 33 runs = 245 files - pass
- Honest status: no pytest suite; recomputation counts as evidence

Thesis Tables 6.3-6.5 - type as Word tables in thesis

R2. Performance Metrics and Known Issues (6.8, 7.3)

- Best accuracy MiniBooNE 0.9376 (ECE 0.010); phoneme 0.8581
- Best OOD mask-0.5 macro AUROC 0.641; regression strong-noise 0.942
- Known issue: 3 collapsed runs (churn/credit/Ailerons) detailed in appendix

Thesis Tables 6.1, 6.2, 6.10 - type as Word tables in thesis

R2. Objectives vs Outcomes - Phase-1 Gate (Table 6.10)

- O1 reproduce 33/33: ACHIEVED | O2 uncertainty table: ACHIEVED
- O3 OOD 7 shifts / 245 files: ACHIEVED | O4 ECE/NLL + 48 plots: ACHIEVED
- Gate decision: Phase 1 complete, proceed to paper drafting

Thesis Table 6.10 - type as Word table in thesis

R2. Phase-2 Plan and Thesis Status (App A)

- Remaining: AUPR/FPR tables, overlay figures, workshop paper draft
- Thesis Chapters 1-5 drafted; Chapter 6 tables ready; appendices stubbed
- Deferred: external baselines, extra-feature retrain, embedding variant

References (IEEE format - all 6 cited works)

- [1] Y. Gorishniy, A. Kotelnikov, and A. Babenko, 'TabM: Advancing tabular deep learning with parameter-efficient ensembling,' in Proc. Int. Conf. Learn. Representations (ICLR), Singapore, 2025. Available: <https://arxiv.org/abs/2410.24210>

- [2] B. Lakshminarayanan, A. Pritzel, and C. Blundell, 'Simple and scalable predictive uncertainty estimation using deep ensembles,' in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 30, Long Beach, CA, USA, 2017, pp. 6402-6413.

- [3] Y. Wen, D. Tran, and J. Ba, 'BatchEnsemble: An alternative approach to efficient ensemble and lifelong learning,' in Proc. Int. Conf. Learn. Representations (ICLR), Addis Ababa, Ethiopia, 2020.

- [4] Y. Gal and Z. Ghahramani, 'Dropout as a Bayesian approximation: Representing model uncertainty in deep learning,' in Proc. Int. Conf. Mach. Learn. (ICML), vol. 48, New York, NY, USA, 2016, pp. 1050-1059.

- [5] D. Hendrycks and K. Gimpel, 'A baseline for detecting misclassified and out-of-distribution examples in neural networks,' in Proc. Int. Conf. Learn. Representations (ICLR), Toulon, France, 2017.

- [6] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, 'On calibration of modern neural networks,' in Proc. Int. Conf. Mach. Learn. (ICML), vol. 70, Sydney, Australia, 2017, pp. 1321-1330.

Thesis References - numbered in order of first citation